Bode Part 2: Sketching and Reading Bode Plots

EGR 345

GVSU

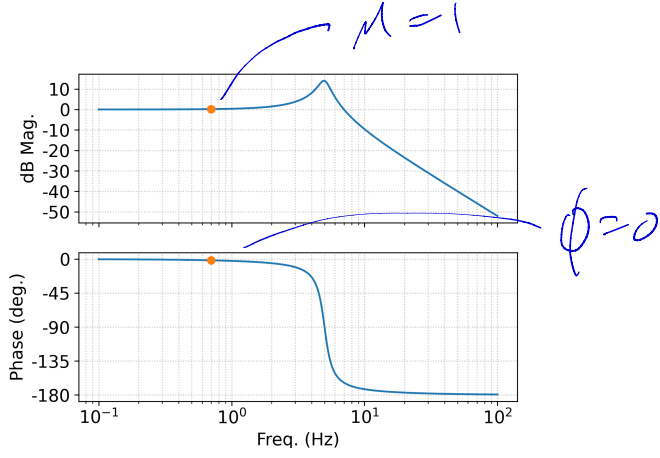
Outline

- quick review: frequency response from Bode
- Bode sketching and reading

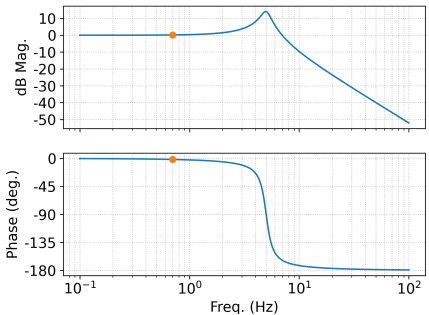
Quick Review

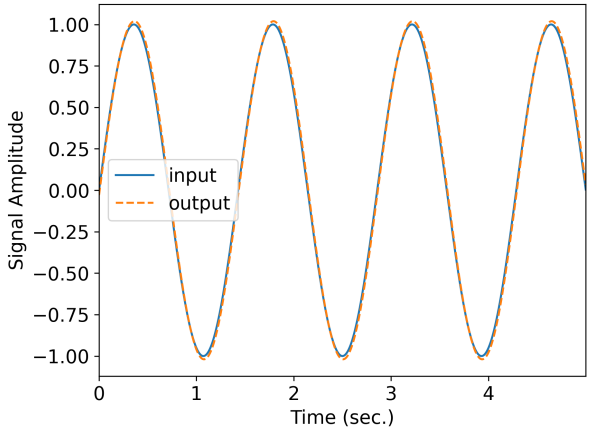
Fixed Sine vs. Bode 1

- what would you expect the steady-state input/output relationship to look like?



Fixed Sine vs. Bode 1 (answer)





0 dB

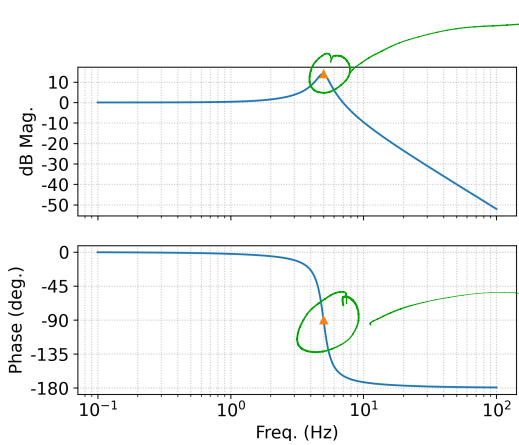
$$20 \log_{10}(M) = 0$$

0 dB

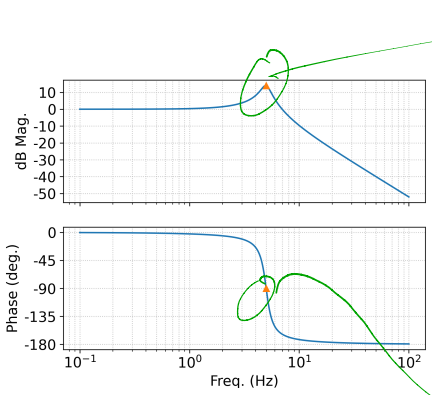
$$M = 10^{0/20} = 1$$

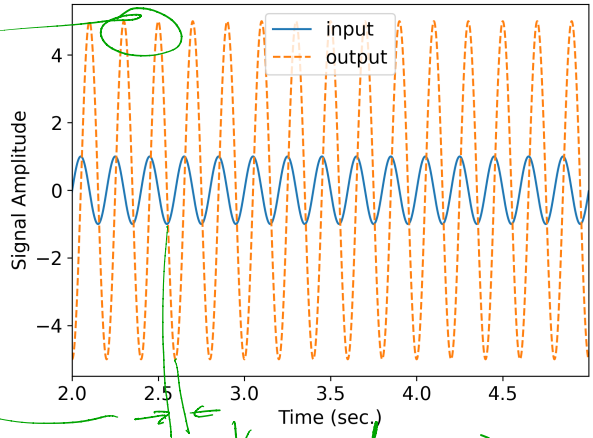
Fixed Sine vs. Bode 2

- what would you expect the steady-state input/output relationship to look like?



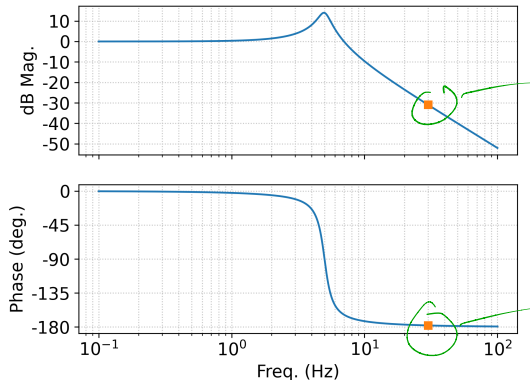
Fixed Sine vs. Bode 2 (answer)



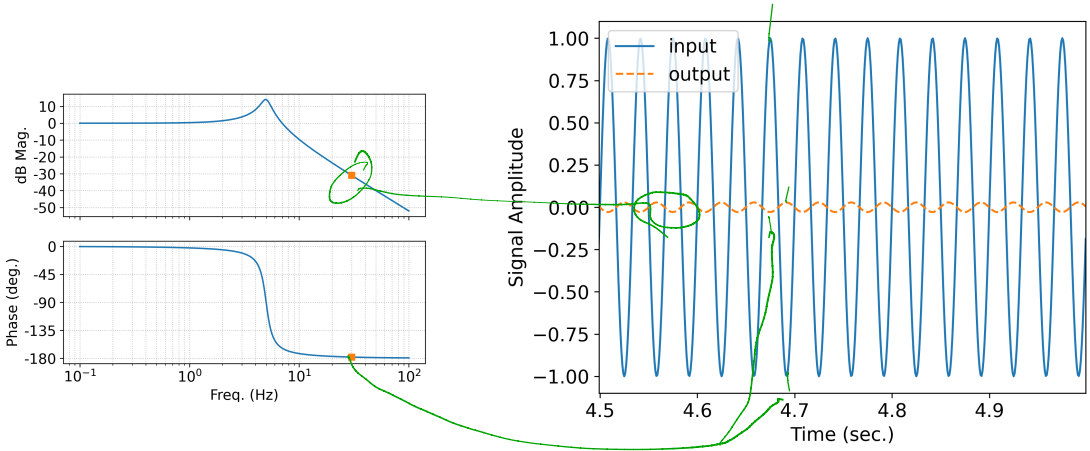


Fixed Sine vs. Bode 3

- what would you expect the steady-state input/output relationship to look like?



Fixed Sine vs. Bode 30 (answer)



Bode Sketching

My claim

Learning to sketch a Bode plot by hand will help you learn to read them.

- forces you to think about the details of the Bode plot
- focuses your thinking on the connections between the terms in the TF and their effects on the Bode plot

Our First Bode Sketching Example

Building block 1: a pure integrator

$$G_1(s) = \frac{2\pi}{s}$$

- multiplication my s is the same as differentiation, so dividing by s is the same as integrating

Bode Plot Table for $G_1(s)$

s is in the denom.
- increasing s by 10x decrease M by 10x

$$G_1(s) = \frac{2\pi}{s}$$

$s = j\omega$

$\frac{1}{0.01j} \cdot \frac{j}{j} = \frac{j}{-0.01} = -100j$

f	ω	s	$G(j\omega)$	M	dB	phase
0.01	$0.01 \cdot 2\pi$	$0.01 \cdot 2\pi j$	$1/(0.01j) = -100j$	100	40	$-90°$
0.1	$0.1 \cdot 2\pi$	$0.1 \cdot 2\pi j$	$1/0.1j = -10j$	10	20	$-90°$
1	2π	$2j\pi$	$1/j = -1j$	1	0	$-90°$
10	20π	$20j\pi$	$1/10j = -0.1j$	0.1	-20	$-90°$
100	200π	$200j\pi$	$1/100 = -0.01j$	0.01	-40	$-90°$
1000	2000π	$2000j\pi$	$1/1000j = -0.001j$	0.001	-60	$-90°$
10000	$20,000\pi$	$20,000j\pi$	$1/10,000j = -0.0001j$	0.0001	-80	-90

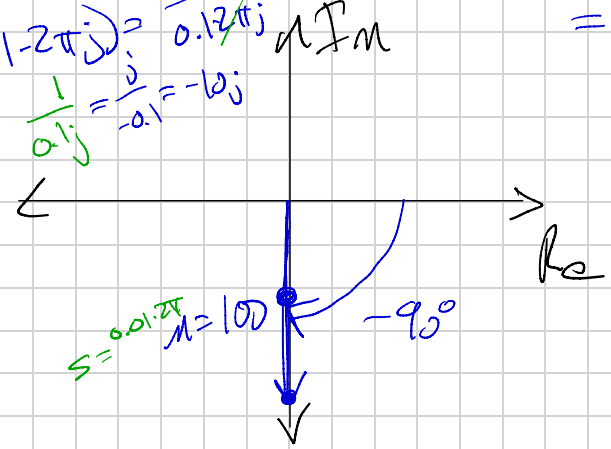
$$G_1(0.01 \cdot 2\pi j) = \frac{2\pi}{0.01(2\pi)j} \quad \frac{1}{0.01j} \cdot \frac{j}{j} = \frac{j}{-0.01}$$

$$= -100j$$

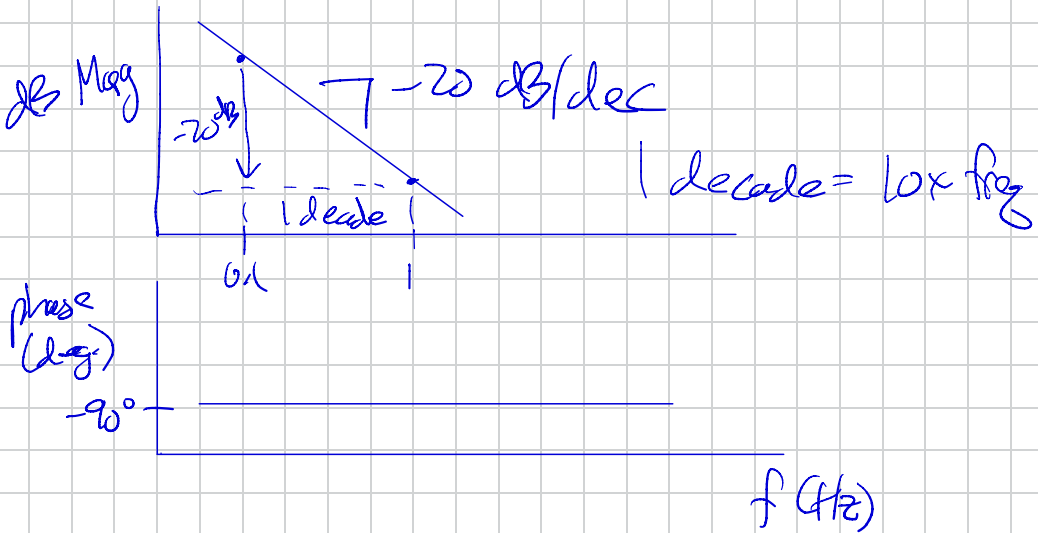
$$G_1(0.1 \cdot 2\pi j) = \frac{2\pi}{0.1 \cdot 2\pi j} \quad \frac{1}{0.1j} = \frac{j}{-0.1} = -10j$$

$M = 10$

$\phi = -90°$



$G_1(s) = \frac{2\pi}{s}$ Pure Integrator



Deductive Reasoning

- Any polynomial with real coefficients can be factored into a product of first and second order terms.
- Therefore, the numerator and denominator of a transfer function can be factored into products of first and second order terms.

Dynamic systems terminology

- The roots of the denominator of a transfer function are called the poles.
- The roots of the numerator of a transfer function are called the zeros.

The six building blocks of transfer functions

- Based on the previous two slides, a transfer function is composed of some product of first and second-order poles along with first and second-order zeros
  - You might think that that leaves us with only four things but poles or zeros at the origin have special names and special significance
- So, the six building blocks are
  1. Pure integrator
  2. Pure differentiator
  3. First order pole
  4. First order zero
  5. Second order poles
  6. Second order zeros

Key Idea

- If we understand how each of the six building blocks affects a Bode plot, then we can sketch the Bode plot of any transfer function.
- If we can recognize the impact of each of the building blocks on a Bode plot, we can infer the transfer function from a Bode plot.

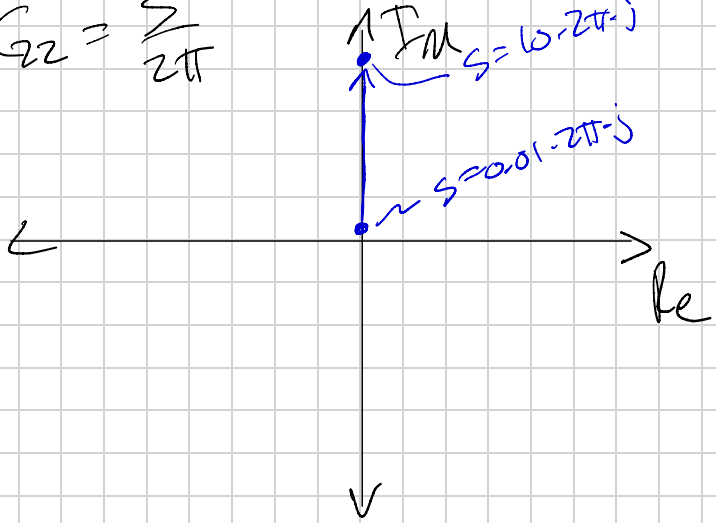
Building Block 2: Pure Diff.

$$G_2(s) = \frac{s}{2\pi} = s$$

- what would happen if s increased by a factor of 10?
- what is the phase? does the phase change with frequency?

f	ω	s	$G(j\omega)$	M	dB	phase
0.01	$0.01 \cdot 2\pi$	$0.01(2\pi j)$	$0.01j$	0.01	-40	90°
0.1	$0.1 \cdot 2\pi$	$0.1(2\pi j)$	$0.1j$	0.1	-20	90°
1	$1 \cdot 2\pi$		j	1	0	90°
10	$10 \cdot 2\pi$		$10j$	10	20	90°
100	$100 \cdot 2\pi$		$100j$	100	40	90°
1000	$1000 \cdot 2\pi$		$1000j$	1000	60	90°
10000	$10,000 \cdot 2\pi$		$10,000j$	10,000	80	90°

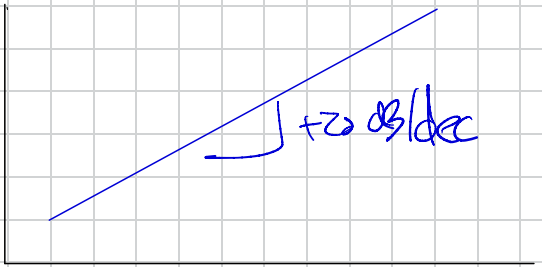
$G_{22} = \frac{S}{2\pi}$

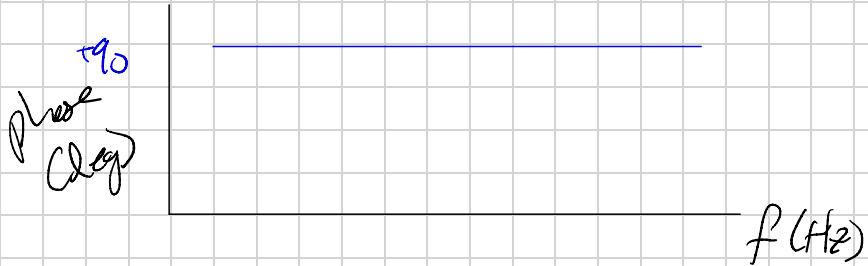


M ↑

$\phi = +90°$

$G_{22} = \frac{s}{2\pi}$





Building block 3: first-order pole

$$G_3(s) = \frac{p}{s+p}$$

- What happens to the magnitude and phase of $G_3(s)$ as the input frequency increases?
- two important frequency ranges:
  - low frequency ($\omega << p$)
  - high frequency ($\omega >> p$)

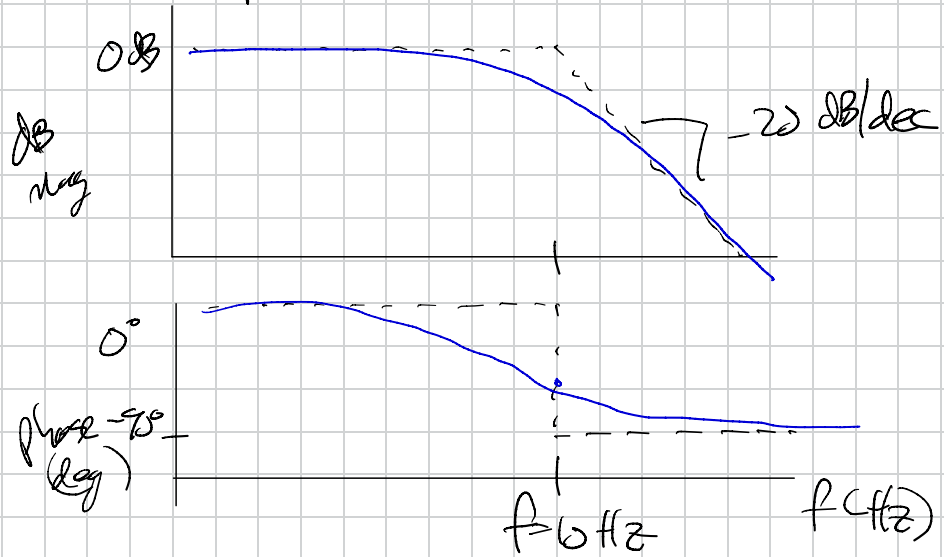
Bode Plot Table for $G_3(s)$

$$G_3(s) = \frac{p}{s+p}$$

assume $p = 10 \cdot 2 \cdot \pi$

f	ω	s	$G(j\omega)$	M	dB	phase
0.01						
0.1						
1						
10						
100						
1000						
10000						

$G_{73}(s) = \frac{p}{s+p}$ $\quad p = 2\pi \cdot 10$



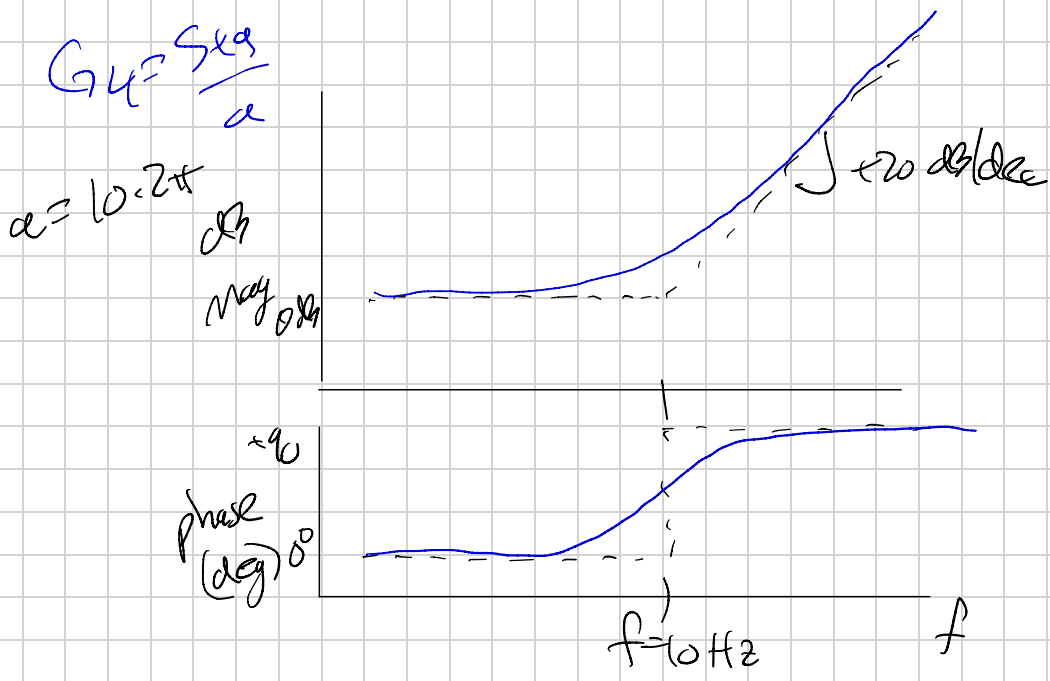
Building Block 4: First Order Zero

$$G_4(s) = \frac{s+a}{a}$$

- two regions:
  - $\omega << a$ — 0 dB, 0 slope, 0° phase
  - $\omega >> a$ ≈ pure diff. → +20 dB/dec slope, +90° phase

$G_4 = \frac{S_{eq}}{a}$

$a = 10 \cdot 2\pi$



Bode Plot Table for $G_4(s)$

$$G_4(s) = \frac{s+a}{a}$$

assume $a = 10 \cdot 2 \cdot \pi$

f	ω	s	$G(j\omega)$	M	dB	phase
0.01						
0.1						
1						
10						
100						
1000						
10000						

Building block 5: second-order (complex) poles

$$G_5(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

- What happens to the magnitude and phase of $G_5(s)$ as the input frequency increases?
- three important frequencies ranges:
  - low frequency ($\omega << \omega_n$)
  - high frequency ($\omega >> \omega_n$)
  - near resonance ($\omega \approx \omega_n$)

Bode Plot Table for $G_5(s)$

$$G_5(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

assume $\omega_n = 10 \cdot 2 \cdot \pi$ and $\zeta = 0.1$

f	ω	s	$G(j\omega)$	M	dB	phase
0.01						
0.1						
1						
10						
100						
1000						
10000						

Building Block 6: Under-Damped Second-Order Zero

$$G_6(s) = \frac{s^2 + 2\zeta\omega_n s + \omega_n^2}{\omega_n^2}$$

- What happens to the magnitude and phase of $G_6(s)$ as the input frequency increases?
- three important frequencies ranges:
  - low frequency ($\omega << \omega_n$)
  - high frequency ($\omega >> \omega_n$)
  - near resonance ($\omega \approx \omega_n$)

Bode Plot Table for $G_6(s)$

$$G_6(s) = \frac{s^2 + 2\zeta\omega_n s + \omega_n^2}{\omega_n^2}$$

assume $\omega_n = 10 \cdot 2 \cdot \pi$ and $\zeta = 0.1$

f	ω	s	$G(j\omega)$	M	dB	phase
0.01						
0.1						
1						
10						
100						
1000						
10000						

Bode Sketching Algorithm

- find all poles and zeros
  - convert them to Hz
  - sort them from lowest to highest frequency
- identify the low frequency behavior
  - pure int, pure diff. or neither
    - first or second order int. or diff. possible
- work your way up in frequency
- when you reach a pole or zero, increment or decrement the slope **and** phase
- continue until you are above the highest frequency pole or zero
- double check the high frequency slope and phase
  - slope: $-20(n-m)$ dB/dec
  - phase: $-90^\circ(n-m)$